

AP Statistics MCQ Practice Exam 07

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AP Statistics MCQ Practice Exam 7

PART A, QUESTIONS

APSTAT-MCQ7-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Ava completes a cycling course in 36 minutes, where cycling times have mean 42 minutes and standard deviation 4 minutes. Noah completes a rowing course in 438 seconds, where rowing times have mean 480 seconds and standard deviation 30 seconds. Lower times indicate better performance. Who performed farther below the mean of the relevant distribution?

- A) Noah, because 42 seconds is a larger raw difference than 6 minutes
- B) Ava, because her z-score is -1.50 compared with Noah's z-score of -1.40
- C) They performed equally well because both times are below their respective means
- D) The performances cannot be compared because the activities use different units

APSTAT-MCQ7-002

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A random sample of 20 warranty claims is classified as approved or not approved. Which value cannot be the sample proportion $\hat{p}$ of approved claims?

- A) 0.15
- B) 0.40
- C) 0.65
- D) 0.33

APSTAT-MCQ7-003

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A county wants to compare the proportions of households that compost food waste in two neighborhoods. Which data-collection plan best supports inference about the difference in household proportions?

- A) Independently select a simple random sample of addresses in each neighborhood and record whether each selected household composts.
- B) Survey shoppers at a farmers market located between the neighborhoods.
- C) Post one online questionnaire and compare respondents who type the two neighborhood names.
- D) Survey several people at each selected address and treat every person's response as an independent household.

APSTAT-MCQ7-004

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A one-sample test of $H_0: \mu = 24$ minutes produces $t = -2.4$. Which interpretation of the test statistic is correct?

- A) The sample mean is 2.4 minutes below 24 minutes.
- B) The population mean has a 2.4% probability of equaling 24 minutes.
- C) The sample mean is 2.4 estimated standard errors below the null mean of 24 minutes.
- D) Exactly 2.4% of the observations are below 24 minutes.

APSTAT-MCQ7-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A regression model predicts equipment temperature y from cooling-airflow setting x . The least-squares slope is -2.4 degrees Celsius per airflow unit. If x increases from 7 to 11, how does the predicted temperature change?

- A) It increases by 9.6 degrees Celsius.
- B) It decreases by 9.6 degrees Celsius.
- C) It decreases by 2.4 degrees Celsius.

D) The change cannot be found without the intercept.

APSTAT-MCQ7-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A company wants to determine whether a redesigned computer interface reduces data-entry errors. Which plan best supports a causal conclusion for the participating employees?

- A) Randomly assign participating employees to the current or redesigned interface, give both groups comparable tasks, and compare error rates.
- B) Let employees choose the interface they prefer and compare their error rates.
- C) Give the redesigned interface to new employees and the current interface to experienced employees.
- D) Ask employees which interface they believe would produce fewer errors.

APSTAT-MCQ7-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a test comparing two population proportions under $H_0: p_1 = p_2$, sample 1 has 42 successes among 60 observations and sample 2 has 30 successes among 50 observations. What is the pooled sample proportion?

- A) 0.600
- B) 0.625
- C) 0.650
- D) 0.655

APSTAT-MCQ7-008

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

For repeated simple random samples of a fixed size from a population with proportion p , the mean of the sampling distribution of $\hat{p}$ equals p . What does this property imply?

- A) Every random sample produces $\hat{p} = p$.
- B) Half of all sample proportions are greater than 0.50.
- C) The estimator $\hat{p}$ has no systematic tendency to overestimate or underestimate p , although individual samples vary.
- D) The standard deviation of $\hat{p}$ is zero.

APSTAT-MCQ7-009

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A researcher constructs a 95% one-sample t confidence interval for μ and tests $H_0: \mu = 50$ using the same data. If the null value is changed to 52 but the sample data and confidence level are unchanged, which result is guaranteed to remain unchanged?

- A) The confidence interval for μ
- B) The t statistic
- C) The p -value
- D) The distance between $\bar{x}$ and the null value

APSTAT-MCQ7-010

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A transit analyst wants to describe the distribution of delays experienced by buses at a particular intersection. Which choice most clearly identifies appropriate observational units and a quantitative variable?

- A) The intersection is the only observational unit, and traffic is labeled light or heavy.
- B) Each randomly selected bus passage is an observational unit, and its delay is measured in seconds relative to schedule.
- C) Each bus route is a variable, and the drivers are classified by route color.
- D) The city's total annual delay is one observational unit measured as delayed or not delayed.

APSTAT-MCQ7-011

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

A laboratory recorded humidity and sensor error together on 30 trials, but someone later sorted the humidity column and error column separately. What is required before a meaningful correlation or regression can be computed?

- A) Convert both sorted columns to z-scores and match equal z-scores.
- B) Pair the smallest humidity with the largest error to maximize the range.
- C) Randomly permute one sorted column and use the resulting pairs.
- D) Recover each trial's original humidity-error pairing; if that pairing cannot be recovered, the association cannot be reconstructed from the marginal lists.

APSTAT-MCQ7-012

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A 95% confidence interval for a population error proportion p is $(0.08, 0.14)$. A performance standard requires $p < 0.10$. Which conclusion is justified?

- A) The standard is definitely met because the interval contains values below 0.10.
- B) The standard is definitely not met because the interval contains values above 0.10.
- C) The interval includes plausible values on both sides of 0.10, so it does not establish whether the standard is met.
- D) Exactly 10% of the population errors lie between 0.08 and 0.14.

APSTAT-MCQ7-013

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A binomial random variable X has mean 12 and standard deviation 3. What are the values of n and p ?

- A) $n = 36$ and $p = 1/3$
- B) $n = 48$ and $p = 1/4$
- C) $n = 16$ and $p = 3/4$
- D) $n = 144$ and $p = 1/12$

APSTAT-MCQ7-014

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

An ordered sample contains 9 observations. Its fifth observation, and therefore its median, is 12; its sixth observation is 15. One new value larger than every original observation is added. What is the median of the enlarged sample?

- A) 12
- B) 15
- C) 27
- D) 13.5

APSTAT-MCQ7-015

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

Researchers compare two short-term treatments for the same quantitative response using a crossover design. Which feature is most important for limiting treatment-order and carryover effects?

- A) Randomize the treatment order for each participant and include an adequate washout period between treatments.
- B) Give every participant treatment A first so the procedure is consistent.
- C) Let participants select which treatment they receive first.
- D) Measure treatment A in the morning and treatment B in the evening for everyone.

APSTAT-MCQ7-016

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A company compares the population false-alarm proportions for a new sensor and an old sensor using independent random samples. It seeks evidence that the new sensor has a lower false-alarm proportion. Which hypotheses match the goal?

- A) $H_0: p_{\text{new}} = p_{\text{old}}$; $H_a: p_{\text{new}} \neq p_{\text{old}}$
- B) $H_0: p_{\text{new}} < p_{\text{old}}$; $H_a: p_{\text{new}} = p_{\text{old}}$
- C) $H_0: p_{\text{new}} = p_{\text{old}}$; $H_a: p_{\text{new}} < p_{\text{old}}$
- D) $H_0: \hat{p}_{\text{new}} = \hat{p}_{\text{old}}$; $H_a: \hat{p}_{\text{new}} < \hat{p}_{\text{old}}$

Questions APSTAT-MCQ7-017 to APSTAT-MCQ7-019 refer to the following stimulus.

The waiting time X , in minutes, for an automated gate has a density curve that is 0 outside $0 \leq X \leq 4$. From $x = 0$ to $x = 4$, the curve is a straight line rising from $(0, 0)$ to $(4, 0.5)$. The total area under the curve is 1.

APSTAT-MCQ7-017

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

The density curve for X is a line from $(0, 0)$ to $(4, 0.5)$. What is $P(X \leq 2)$?

- A) 0.125
- B) 0.500
- C) 0.750
- D) 0.250

APSTAT-MCQ7-018

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

The density curve for X is a line from $(0, 0)$ to $(4, 0.5)$. What is $P(X > 3)$?

- A) $5/16$
- B) $7/16$
- C) $9/16$
- D) $3/4$

APSTAT-MCQ7-019

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

The density curve for X is a line from $(0, 0)$ to $(4, 0.5)$. Which value is approximately the median waiting time, with half the area to its left and half to its right?

- A) 2.83 minutes
- B) 2.00 minutes
- C) 3.46 minutes
- D) 1.41 minutes

APSTAT-MCQ7-020

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A distribution of error scores is right-skewed with median 12 and IQR 6. An accuracy score is defined as 100 minus the error score. Which description of the accuracy-score distribution is correct?

- A) Right-skewed with median 88 and IQR 94
- B) Symmetric with median 100 and IQR 6
- C) Left-skewed with median 88 and IQR 6
- D) Left-skewed with median -12 and IQR -6

APSTAT-MCQ7-021

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A researcher duplicates every row of a dataset and then reruns a one-sample t -test. Software reports a smaller standard error and p -value because the displayed sample size doubled. Which assessment is correct?

- A) The new result is more precise because every copied row counts as a new independent observation.
- B) The apparent gain in precision is invalid because copying observations adds no independent information.
- C) The result is valid only if the original sample mean was zero.
- D) Duplicating rows changes the population parameter being tested.

APSTAT-MCQ7-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Two independent samples each have size 400. Sample A has $\hat{p}_A = 0.10$, and sample B has $\hat{p}_B = 0.50$. Separate 95% confidence intervals are constructed for the two population proportions using the standard one-proportion z-interval. Which interval has the smaller margin of error?

- A) Sample A's interval, because $0.10(0.90)$ is less than $0.50(0.50)$
- B) Sample B's interval, because 0.50 is farther from 0
- C) The margins are equal because the sample sizes and confidence levels are equal
- D) The margins cannot be compared without knowing the population sizes

APSTAT-MCQ7-023

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

An empirical cumulative relative-frequency graph for customer wait times has value 0.72 at 30 seconds. What does this mean?

- A) The mean wait time is 30 seconds for 72% of customers.
- B) Exactly 28% of wait times equal 30 seconds.
- C) The standard deviation of wait times is 0.72 second.
- D) Approximately 72% of observed wait times are at most 30 seconds.

APSTAT-MCQ7-024

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Software output for an unpooled two-sample t procedure reports $df = 17.6$. A student claims that degrees of freedom must always be an integer. Which assessment is correct?

- A) The output is invalid because df must equal the smaller sample size.
- B) The output should always use $df = n_1 + n_2 - 2$.
- C) The Welch approximation can produce noninteger df , and software may use 17.6 directly.

D) A noninteger df proves that the observations are paired.

APSTAT-MCQ7-025

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

A spinner lands blue with probability 0.20, green with probability 0.30, and gold with probability 0.50. Which assignment of random digits 0 through 9 correctly simulates one spin?

- A) 0–1 blue, 2–4 green, 5–9 gold
- B) 0–2 blue, 3–5 green, 6–9 gold
- C) 0 blue, 1–3 green, 4–9 gold
- D) 0–1 blue, 2–5 green, 6–9 gold

APSTAT-MCQ7-026

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A probability sample for a mailed public-library survey has a low response rate. Which follow-up plan is most useful for assessing possible nonresponse bias?

- A) Replace every nonrespondent with the nearest person listed in the directory.
- B) Increase the reported sample size by counting each respondent twice.
- C) Compare early respondents with later respondents only and assume they represent all nonrespondents.
- D) Randomly select a subset of nonrespondents for intensive follow-up and compare their answers with those of initial respondents.

APSTAT-MCQ7-027

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Under a null hypothesis of no treatment effect, 5,000 random reassignments are generated. In 73 reassignments, the absolute difference in sample proportions is at least as large as the observed

absolute difference. What is the randomization-test p-value?

- A) 0.0073
- B) 0.0146
- C) 0.0730
- D) 0.1460

APSTAT-MCQ7-028

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Two laboratory instruments measure the same concentration. Their readings have correlation $r = 0.98$, but the new instrument consistently reads about 5 units higher than the old instrument. Which conclusion is most appropriate?

- A) The instruments agree closely because a correlation near 1 guarantees equal readings.
- B) The new instrument is more precise because its readings are larger.
- C) The readings have a strong linear association, but high correlation does not rule out systematic disagreement.
- D) Subtracting 5 from the new readings would make the correlation equal to 0.

APSTAT-MCQ7-029

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For two independent populations, an estimate of $\mu_1 - \mu_2$ is 2.0 units with standard error 1.0 unit. What is the t statistic for testing $H_0: \mu_1 - \mu_2 = 5.0$?

- A) 7.0
- B) 3.0
- C) -2.0
- D) -3.0

APSTAT-MCQ7-030

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A normal distribution has standard deviation 8. The z-scores of its first and third quartiles are approximately -0.674 and 0.674 . What is the IQR of the distribution?

- A) Approximately 10.8
- B) Approximately 5.4
- C) Approximately 8.0
- D) Approximately 16.0

APSTAT-MCQ7-031

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A bike-share agency wants to estimate the probability distribution of the daily number of rentals over an entire year. Which data-collection plan is most appropriate?

- A) Record only the ten days with the largest numbers of rentals.
- B) Use every day in July because summer has the greatest activity.
- C) Select dates at random from throughout the full year and record the number of rentals on each selected date.
- D) Ask riders to report how many rentals they believe occur on a typical day.

APSTAT-MCQ7-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

For a right-tailed one-proportion z-test of $H_0: p = 0.50$ with $n = 100$, reject H_0 when $z > 1.645$. What is the smallest number of successes that will lead to rejection?

- A) 58
- B) 59
- C) 60
- D) 67

APSTAT-MCQ7-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A food scientist studies the effects of oven temperature (low or high) and convection fan setting (off or on) on crust crispness. Which design best permits separate assessment of both factors and their combination?

- A) Randomly assign similar loaves to all four temperature-by-fan treatment combinations, with replication in each combination.
- B) Use low temperature only when the fan is off and high temperature only when the fan is on.
- C) Let bakers choose one of the four combinations for each loaf.
- D) Apply the four combinations on four different days with no random assignment or replication.

Questions APSTAT-MCQ7-034 to APSTAT-MCQ7-036 refer to the following stimulus.

Three calibration trials produced the paired data below. A least-squares line predicts response y from intensity setting x .

x : 1 2 3

y : 2 4 3

APSTAT-MCQ7-034

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For the paired data (1, 2), (2, 4), and (3, 3), what is the least-squares regression line predicting y from x ?

- A) $\hat{y} = 3 + 2x$
- B) $\hat{y} = 3 - 0.5x$
- C) $\hat{y} = 1 + x$
- D) $\hat{y} = 2 + 0.5x$

APSTAT-MCQ7-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For the paired data (1, 2), (2, 4), and (3, 3), the least-squares line is $\hat{y} = 2 + 0.5x$. What is the residual for the trial with $x = 3$?

- A) 0.5
- B) 3.5
- C) -0.5
- D) -3.5

APSTAT-MCQ7-036

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

For the paired data (1, 2), (2, 4), and (3, 3), the least-squares line is $\hat{y} = 2 + 0.5x$. Which trial is fitted least closely in the vertical direction?

- A) The trial with $x = 1$, because its residual is -0.5
- B) The trial with $x = 2$, because its residual is 1 and has the largest absolute value
- C) The trial with $x = 3$, because the intensity setting is largest
- D) All three trials are fitted equally closely because they determine the line

APSTAT-MCQ7-037

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

Participants are randomly assigned to rehabilitation program A or B, and each participant is measured before and after the program. Researchers want to compare the population mean changes for the two programs. Which analysis matches the goal?

- A) A one-proportion z-test using the number who improve
- B) A paired t-test pairing one participant from program A with one from program B
- C) Compute each participant's after-minus-before change, then use a two-sample t procedure to compare mean changes between programs.
- D) A one-sample t procedure using all before measurements only

APSTAT-MCQ7-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A random voucher credit X can equal 0, 10, or 20 dollars. Its probabilities are $P(X = 0) = 0.50$, $P(X = 10) = q$, and $P(X = 20) = 0.50 - q$. If $E(X) = 6$ dollars, what are q and $P(X = 20)$?

- A) $q = 0.40$ and $P(X = 20) = 0.10$
- B) $q = 0.10$ and $P(X = 20) = 0.40$
- C) $q = 0.50$ and $P(X = 20) = 0$
- D) $q = 0.30$ and $P(X = 20) = 0.20$

APSTAT-MCQ7-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a sample of $n = 200$, $\hat{p} = 0.30$. For a confidence interval, the estimated standard error uses $\hat{p}$. For testing $H_0: p = 0.20$, the null standard error uses $p_0 = 0.20$. Which pair of standard errors is correct?

- A) 0.0283 for the interval and 0.0324 for the test
- B) 0.0300 for both procedures
- C) 0.0324 for both procedures
- D) 0.0324 for the interval and 0.0283 for the test

APSTAT-MCQ7-040

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A right-tailed test of $H_0: p = 0.50$ is designed to have power 0.80 when $p = 0.60$. With the same sample size and significance level, what should generally happen to power if the true proportion is $p = 0.70$?

- A) It should fall below 0.50 because 0.70 is farther from the null value.
- B) It should be greater than 0.80 because 0.70 is farther into the direction specified by H_a .
- C) It must remain exactly 0.80 for every value in H_a .
- D) It becomes the significance level because H_0 is false.

APSTAT-MCQ7-041

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A city wants to estimate how the proportion of commuters using bicycles changes between this year and four years from now. Which plan best supports comparison of the city populations at the two times?

- A) Take an independent probability sample of city commuters at each time using comparable questions and sampling frames.
- B) Survey only current bicycle commuters and contact the same volunteers four years later.
- C) Compare responses from two open social-media polls conducted four years apart.
- D) Ask current commuters to predict how they will travel four years from now.

APSTAT-MCQ7-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Two samples have the same size. The median of sample A is 10, and the median of sample B is 20. What can be concluded about the median of the combined sample?

- A) It must be 10.
- B) It must be 15.
- C) It cannot be determined from the two sample medians alone.
- D) It must be 20.

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ7-001, Correct: B, Correct Answer: Ava, because her z-score is -1.50 compared with Noah's z-score of -1.40 , Explanation: Ava's standardized time is $(36 - 42)/4 = -1.50$, and Noah's is $(438 - 480)/30 = -1.40$. Because lower times are better, Ava performed slightly farther in the favorable direction relative to her distribution.

APSTAT-MCQ7-002, Correct: D, Correct Answer: 0.33, Explanation: With $n = 20$, $\hat{p} = X/20$ for an integer count X , so possible proportions occur in increments of 0.05. The values 0.15, 0.40, and 0.65 correspond to 3, 8, and 13 approvals, but 0.33 does not correspond to an integer count.

APSTAT-MCQ7-003, Correct: A, Correct Answer: Independently select a simple random sample of addresses in each neighborhood and record whether each selected household composts., Explanation: The household is the observational unit and the response is binary. Independent probability samples from both neighborhoods support inference about the difference in their population proportions.

APSTAT-MCQ7-004, Correct: C, Correct Answer: The sample mean is 2.4 estimated standard errors below the null mean of 24 minutes., Explanation: A t statistic measures the signed distance from the null value in estimated standard-error units. The negative sign indicates $\bar{x} < 24$, and the magnitude 2.4 gives the standardized distance, not a raw time or probability.

APSTAT-MCQ7-005, Correct: B, Correct Answer: It decreases by 9.6 degrees Celsius., Explanation: A slope of -2.4 means each one-unit increase in airflow changes predicted temperature by -2.4 degrees. The four-unit increase gives $\Delta\hat{y} = -2.4(11 - 7) = -9.6$ degrees Celsius.

APSTAT-MCQ7-006, Correct: A, Correct Answer: Randomly assign participating employees to the current or redesigned interface, give both groups comparable tasks, and compare error rates., Explanation: Random assignment tends to balance other employee characteristics across the interface groups, and comparable tasks prevent task difficulty from being confounded with interface type.

APSTAT-MCQ7-007, Correct: D, Correct Answer: 0.655, Explanation: Under H_0 : $p_1 = p_2$, pool the successes and observations: $\hat{p}_{\text{pool}} = (42 + 30)/(60 + 50) = 72/110 \approx 0.655$.

APSTAT-MCQ7-008, Correct: C, Correct Answer: The estimator $\hat{p}$ has no systematic tendency to overestimate or underestimate p , although individual samples vary., Explanation: An estimator whose sampling-distribution mean equals the parameter is unbiased. Unbiasedness is a repeated-sampling property and does not require every estimate to equal p .

APSTAT-MCQ7-009, Correct: A, Correct Answer: The confidence interval for μ , Explanation: A confidence interval is determined by the sample data, confidence level, and procedure; it does not depend on which null value is later tested. Changing the null value changes $\bar{x} - \mu_0$ and therefore generally changes the t statistic and p -value.

APSTAT-MCQ7-010, Correct: B, Correct Answer: Each randomly selected bus passage is an observational unit, and its delay is measured in seconds relative to schedule., Explanation: Each bus passage supplies one observation, and delay in seconds is a clearly defined quantitative variable whose distribution can be described.

APSTAT-MCQ7-011, Correct: D, Correct Answer: Recover each trial's original humidity-error pairing; if that pairing cannot be recovered, the association cannot be reconstructed from the marginal lists., Explanation: Correlation and regression depend on which x -value occurred with each y -value. Separately sorted marginal lists preserve individual distributions but destroy the joint pairings that determine association.

APSTAT-MCQ7-012, Correct: C, Correct Answer: The interval includes plausible values on both sides of 0.10, so it does not establish whether the standard is met., Explanation: Because 0.10 lies inside the interval, the data are compatible with a population error proportion both below and above the performance threshold. The interval therefore does not resolve compliance at the stated confidence level.

APSTAT-MCQ7-013, Correct: B, Correct Answer: $n = 48$ and $p = 1/4$, Explanation: For a binomial variable, $\mu = np = 12$ and $\sigma^2 = np(1 - p) = 9$. Thus $1 - p = 9/12 = 3/4$, so $p = 1/4$ and $n = 12/(1/4) = 48$.

APSTAT-MCQ7-014, Correct: D, Correct Answer: 13.5, Explanation: After a new largest value is added, there are 10 observations. The middle positions are the fifth and sixth original observations, so the new

median is $(12 + 15)/2 = 13.5$.

APSTAT-MCQ7-015, Correct: A, Correct Answer: Randomize the treatment order for each participant and include an adequate washout period between treatments., Explanation: Randomizing order balances systematic order effects, and a washout period reduces carryover from the first treatment into the second treatment period.

APSTAT-MCQ7-016, Correct: C, Correct Answer: $H_0: p_{\text{new}} = p_{\text{old}}$; $H_a: p_{\text{new}} < p_{\text{old}}$, Explanation: Hypotheses use population parameters, equality belongs in H_0 , and the claim that the new false-alarm proportion is lower determines a left-tailed alternative.

APSTAT-MCQ7-017, Correct: D, Correct Answer: 0.250, Explanation: At $x = 2$, the line's height is 0.25. The probability is the triangular area from 0 to 2: $(1/2)(2)(0.25) = 0.250$.

APSTAT-MCQ7-018, Correct: B, Correct Answer: $7/16$, Explanation: The triangular area to the left of 3 is $(1/2)(3)(0.375) = 9/16$. Therefore $P(X > 3) = 1 - 9/16 = 7/16$.

APSTAT-MCQ7-019, Correct: A, Correct Answer: 2.83 minutes, Explanation: At $x = m$, the density height is $m/8$, so the triangular area to the left is $(1/2)m(m/8) = m^2/16$. Setting this area equal to 0.5 gives $m = \sqrt{8} \approx 2.83$ minutes.

APSTAT-MCQ7-020, Correct: C, Correct Answer: Left-skewed with median 88 and IQR 6, Explanation: The transformation $y = 100 - x$ reflects the distribution and shifts it, so right skew becomes left skew. The median becomes $100 - 12 = 88$, and multiplying by -1 leaves the IQR's magnitude unchanged at 6.

APSTAT-MCQ7-021, Correct: B, Correct Answer: The apparent gain in precision is invalid because copying observations adds no independent information., Explanation: The copied rows are perfectly dependent duplicates rather than new observations. Treating them as independent artificially inflates n and understates the true standard error.

APSTAT-MCQ7-022, Correct: A, Correct Answer: Sample A's interval, because $0.10(0.90)$ is less than $0.50(0.50)$, Explanation: With the same n and z^* , margin of error is proportional to $\sqrt{[\hat{p}(1 - \hat{p})]}$. This product is 0.09 for sample A and 0.25 for sample B, so sample A's interval is narrower.

APSTAT-MCQ7-023, Correct: D, Correct Answer: Approximately 72% of observed wait times are at most 30 seconds., Explanation: An empirical cumulative relative frequency at 30 seconds is the proportion of observations less than or equal to 30 seconds.

APSTAT-MCQ7-024, Correct: C, Correct Answer: The Welch approximation can produce noninteger df, and software may use 17.6 directly., Explanation: The unpooled two-sample t procedure uses an approximate degrees-of-freedom calculation that need not be an integer. Statistical software can evaluate the t distribution at the reported decimal df.

APSTAT-MCQ7-025, Correct: A, Correct Answer: 0–1 blue, 2–4 green, 5–9 gold, Explanation: The assignment gives 2 of 10 digits to blue, 3 of 10 to green, and 5 of 10 to gold, matching probabilities 0.20, 0.30, and 0.50.

APSTAT-MCQ7-026, Correct: D, Correct Answer: Randomly select a subset of nonrespondents for intensive follow-up and compare their answers with those of initial respondents., Explanation: Intensive

follow-up can reveal whether people who initially failed to respond differ systematically from respondents. Convenience replacement or double-counting cannot diagnose the bias.

APSTAT-MCQ7-027, Correct: B, Correct Answer: 0.0146, Explanation: The simulation already counts results at least as extreme in either direction by using the absolute difference. Therefore the estimated p-value is $73/5,000 = 0.0146$, with no additional doubling.

APSTAT-MCQ7-028, Correct: C, Correct Answer: The readings have a strong linear association, but high correlation does not rule out systematic disagreement., Explanation: Correlation measures the strength of linear association, not equality or agreement. Adding a constant to one instrument's readings can preserve r while producing a systematic offset.

APSTAT-MCQ7-029, Correct: D, Correct Answer: -3.0 , Explanation: The statistic compares the estimated difference with the null difference: $t = [2.0 - 5.0]/1.0 = -3.0$.

APSTAT-MCQ7-030, Correct: A, Correct Answer: Approximately 10.8, Explanation: The IQR is the distance from Q1 to Q3: $[0.674 - (-0.674)](8) = 1.348(8) = 10.784$, approximately 10.8.

APSTAT-MCQ7-031, Correct: C, Correct Answer: Select dates at random from throughout the full year and record the number of rentals on each selected date., Explanation: Random selection across the entire year represents seasonal and weekday variation in the target distribution. Selecting only busy or summer days would bias the distribution upward.

APSTAT-MCQ7-032, Correct: B, Correct Answer: 59, Explanation: Under H_0 , $SE_{\hat{p}} = \sqrt{[0.50(0.50)/100]} = 0.05$. Rejection requires $(\hat{p} - 0.50)/0.05 > 1.645$, so $\hat{p} > 0.58225$. The smallest attainable proportion above that cutoff is $59/100$, corresponding to 59 successes.

APSTAT-MCQ7-033, Correct: A, Correct Answer: Randomly assign similar loaves to all four temperature-by-fan treatment combinations, with replication in each combination., Explanation: A replicated factorial design varies both factors in all combinations. Random assignment permits fair comparisons of temperature, fan setting, and how their effects may work together.

APSTAT-MCQ7-034, Correct: D, Correct Answer: $\hat{y} = 2 + 0.5x$, Explanation: Here $\bar{x} = 2$ and $\bar{y} = 3$. The slope is $\Sigma[(x - 2)(y - 3)]/\Sigma(x - 2)^2 = 1/2 = 0.5$, and the intercept is $3 - 0.5(2) = 2$.

APSTAT-MCQ7-035, Correct: C, Correct Answer: -0.5 , Explanation: At $x = 3$, $\hat{y} = 2 + 0.5(3) = 3.5$. The residual is observed minus predicted, $3 - 3.5 = -0.5$.

APSTAT-MCQ7-036, Correct: B, Correct Answer: The trial with $x = 2$, because its residual is 1 and has the largest absolute value, Explanation: The residuals for $x = 1, 2$, and 3 are $-0.5, 1$, and -0.5 . The $x = 2$ trial has the greatest vertical distance from the line and the largest squared residual contribution.

APSTAT-MCQ7-037, Correct: C, Correct Answer: Compute each participant's after-minus-before change, then use a two-sample t procedure to compare mean changes between programs., Explanation: Each participant supplies one change score, preserving the within-person before/after pairing. The program A and B groups are independent, so their mean change scores are compared with a two-sample t procedure.

APSTAT-MCQ7-038, Correct: A, Correct Answer: $q = 0.40$ and $P(X = 20) = 0.10$, Explanation: The mean condition gives $10q + 20(0.50 - q) = 6$. Thus $10 - 10q = 6$, so $q = 0.40$ and $P(X = 20) = 0.50 - 0.40 = 0.10$.

APSTAT-MCQ7-039, Correct: D, Correct Answer: 0.0324 for the interval and 0.0283 for the test, Explanation: For the interval, $SE = \sqrt{[0.30(0.70)/200]} \approx 0.0324$. Under H_0 , the test uses $SE_0 = \sqrt{[0.20(0.80)/200]} \approx 0.0283$.

APSTAT-MCQ7-040, Correct: B, Correct Answer: It should be greater than 0.80 because 0.70 is farther into the direction specified by H_a ., Explanation: With n and α fixed, values farther from the null in the alternative direction are more likely to fall in the rejection region, so power increases.

APSTAT-MCQ7-041, Correct: A, Correct Answer: Take an independent probability sample of city commuters at each time using comparable questions and sampling frames., Explanation: Comparable probability samples at both times represent the changing commuter populations. Tracking only current bicycle commuters or using voluntary polls would not estimate the two citywide proportions fairly.

APSTAT-MCQ7-042, Correct: C, Correct Answer: It cannot be determined from the two sample medians alone., Explanation: The combined middle positions depend on how all observations from the two samples interleave. Equal sample sizes and the two separate medians do not determine those positions.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ7

· APSTAT-MCQ7-001 to 005: Unit 1 cycling and rowing completion times compare favorable negative z-scores across different scales; Unit 2 a binary sample of size 20 restricts $\hat{p}$ to multiples of 0.05; Unit 3 household composting proportions use independent address SRSs from two neighborhoods; Unit 4 interprets $t = -2.4$ as a sample mean 2.4 estimated standard errors below $\mu_0 = 24$; Unit 5 uses a slope of -2.4 degrees per airflow unit to find the predicted change over a four-unit airflow increase.

Misconceptions include comparing raw deviations across units, allowing a noninteger success count, sampling convenient individuals instead of households, reading t as raw minutes or probability, and treating slope as the total change or requiring the intercept for a predicted difference.

· APSTAT-MCQ7-006 to 010: Unit 1 a current-versus-redesigned interface experiment randomly assigns employees and holds task difficulty comparable; Unit 3 sensor-test samples pool 72 successes among 110 observations under equal proportions; Unit 2 unbiasedness of $\hat{p}$ is interpreted as lack of systematic repeated-sampling error rather than exactness; Unit 4 changing a tested null mean while holding sample data fixed leaves a confidence interval unchanged but alters the t statistic; Unit 1 bus passages and delay seconds identify observational units and a quantitative variable. Misconceptions include self-selection into treatments, averaging sample proportions instead of pooling counts, equating unbiased with invariant, building a data-based interval around the null value, and treating an aggregate city total as a distribution of observations.

· APSTAT-MCQ7-011 to 015: Unit 5 independently sorted humidity and sensor-error columns show that correlation and regression require recovery of the original trial pairings; Unit 3 a 95% error-proportion interval (0.08, 0.14) straddles a 0.10 performance threshold and cannot establish compliance; Unit 2 reverses binomial mean 12 and standard deviation 3 to obtain $p = 1/4$ and $n = 48$; Unit 1 adding a new maximum to nine ordered observations moves the median to the average of the old fifth and sixth values; Unit 4 a crossover treatment comparison randomizes order and uses washout to limit carryover. Misconceptions include reconstructing association from marginal ranks, declaring compliance from only one side of an interval, confusing variance with standard deviation, leaving the odd-sample median unchanged after n becomes even, and using a fixed treatment order.

- APSTAT-MCQ7-016 to 020: Unit 3 new-versus-old sensor false-alarm proportions formulate a left-tailed population-parameter alternative; Unit 2 an increasing triangular gate-wait density set uses geometric area for $P(X \leq 2)$, a complementary tail $P(X > 3)$, and the median from half the area; Unit 1 reflecting right-skewed error scores through accuracy = 100 – error reverses skew while preserving IQR magnitude. Misconceptions include putting an inequality in H_0 or using statistics in hypotheses, treating density height as probability, subtracting the wrong triangular area, locating the median at the midpoint of the support, and allowing a negative IQR.
- APSTAT-MCQ7-021 to 025: Unit 4 duplicated data rows create pseudoreplication and an artificially small one-mean standard error; Unit 3 equal-size one-proportion intervals at $\hat{p} = 0.10$ and 0.50 compare $\hat{p}(1 - \hat{p})$ and width; Unit 1 an empirical cumulative-frequency curve value 0.72 at 30 seconds means 72% of waits are at most 30 seconds; Unit 4 Welch's unpooled two-sample t approximation legitimately reports noninteger $df = 17.6$; Unit 2 a three-color spinner maps probabilities 0.20 , 0.30 , and 0.50 to 2, 3, and 5 random digits. Misconceptions include counting copies as independent information, assuming equal n forces equal margin, reading a cumulative value as a point mass or mean, forcing a pooled integer df , and assigning the wrong number of digits to outcomes.
- APSTAT-MCQ7-026 to 030: Unit 1 a low-response library mail survey uses intensive follow-up of a random subset of nonrespondents to assess bias; Unit 3 a two-sided randomization test counts 73 of 5,000 absolute differences for $p = 0.0146$; Unit 5 two instruments with $r = 0.98$ but a five-unit offset distinguish association from agreement; Unit 4 tests a nonzero null difference in means using $(2 - 5)/1 = -3$; Unit 1 a normal distribution uses supplied quartile z -scores and $\sigma = 8$ to calculate $IQR \approx 10.8$. Misconceptions include convenience replacement, doubling a p -value already based on absolute differences, treating high correlation as agreement, omitting the null difference from a t numerator, and using one quartile displacement instead of both.
- APSTAT-MCQ7-031 to 035: Unit 2 randomly sampled dates across a full year estimate a bike-rental count distribution; Unit 3 a right-tailed test with $n = 100$ and critical $z = 1.645$ converts its cutoff to a minimum of 59 successes; Unit 1 a replicated 2×2 oven-temperature and convection-fan factorial experiment includes all four randomized combinations; Unit 5 a three-trial calibration stimulus derives $\hat{y} = 2 + 0.5x$ from raw pairs and calculates the $x = 3$ residual as -0.5 . Misconceptions include selecting only high-activity dates, failing to round an attainable sample proportion to the next integer count, confounding two factors by moving them together, using an arbitrary line rather than least squares, and reversing residual sign.
- APSTAT-MCQ7-036 to 040: Unit 5 the calibration stimulus compares absolute vertical residuals and identifies the $x = 2$ trial as least closely fitted; Unit 4 a two-program pre/post design computes individual changes before applying a two-sample t procedure; Unit 2 a voucher-credit distribution uses normalization and $E(X) = 6$ to solve $q = 0.40$ and the remaining probability 0.10 ; Unit 3 distinguishes the $\hat{p}$ -based interval standard error from the p_0 -based one-proportion test standard error; Unit 3 power for $H_a: p > 0.50$ increases when the true value moves from 0.60 to 0.70 . Misconceptions include choosing the largest x instead of largest residual, pairing unrelated participants across programs, solving the probability total without using the mean condition, using one standard error for both procedures, and assuming power is constant throughout H_a .
- APSTAT-MCQ7-041 to 042: Unit 1 repeated cross-sectional probability samples compare city bicycle-commuting proportions four years apart; Unit 1 equal-size samples with medians 10 and 20 demonstrate that the combined median depends on the full interleaving of observations. Misconceptions include following only current cyclists or using voluntary polls, and averaging subgroup medians without the raw order information.
- Unit counts for APSTAT-MCQ7: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: increasing triangular gate-wait density, APSTAT-MCQ7-017 to 019; raw three-trial intensity-and-response regression, APSTAT-MCQ7-034 to 036.